

# Stability

## Chapter 6

## Routh Hurwitz Criterion: Special Case

### 2. Entire row is zero

An entire row consists of zeros because there is an even polynomial that is a factor of the original polynomial.

## Example#6.4

Determine the number of right-half-plane poles in the closed loop transfer function  $T(s)$

$$T(s) = \frac{10}{s^5 + 7s^4 + 6s^3 + 42s^2 + 8s + 56}$$

Routh table

|       |                |                 |                 |
|-------|----------------|-----------------|-----------------|
| $s^5$ | 1              | 6               | 8               |
| $s^4$ | <del>7</del> 1 | <del>42</del> 6 | <del>56</del> 8 |
| $s^3$ | 0              | 0               | 0               |
| $s^2$ |                |                 |                 |
| $s^1$ |                |                 |                 |
| $s^0$ |                |                 |                 |

## Procedure:

- Create an auxiliary polynomial with the row immediately above the row of zeros.
- Start with the power of “s” in the index column and include alternate powers thereafter

$$P(s) = s^4 + 6s^2 + 8 \quad \text{Order of even polynomial}=4$$

- Differentiate the polynomial P(s) with respect to “s”

$$\frac{dP(s)}{ds} = 4s^3 + 12s + 0$$

- Use these coefficients to replace the row of zeros
- Remainder of the table is formed in a straightforward manner by following the standard form

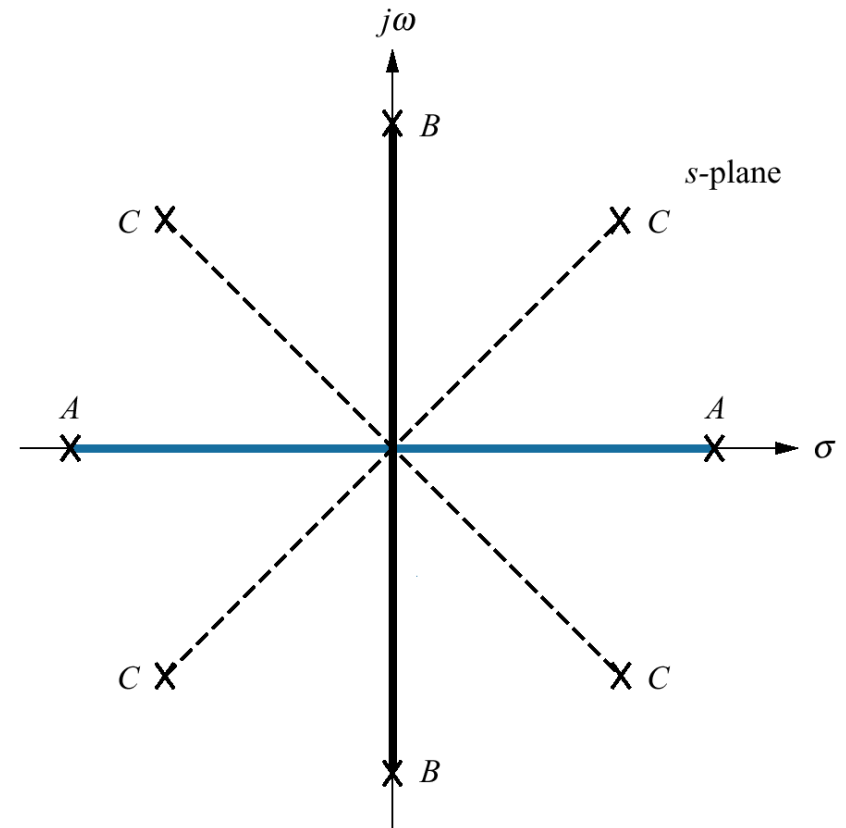
## Example: 6.4 Routh Table

|       |                             |                              |                             |
|-------|-----------------------------|------------------------------|-----------------------------|
| $s^5$ | 1                           | 6                            | 8                           |
| $s^4$ | <del>7</del> 1              | <del>42</del> 6              | <del>56</del> 8             |
| $s^3$ | <del>0</del> <del>4</del> 1 | <del>0</del> <del>12</del> 3 | <del>0</del> <del>0</del> 0 |
| $s^2$ | 3                           | 8                            | 0                           |
| $s^1$ | $\frac{1}{3}$               | 0                            | 0                           |
| $s^0$ | 8                           | 0                            | 0                           |

Conclusion: No sign change. So no RHP poles  
 What about jw axis poles????

## Row of zero

- Existence of even polynomial that is a factor of the original polynomial
- Even polynomials have roots only symmetrically placed about the origin
- The row previous to the zero row contain this even polynomial
- Everything from the row containing the even polynomial downward is a test only for the even polynomial



A: Real and symmetrical about the origin  
B: Imaginary and symmetrical about the origin  
C: Quadrantal and symmetrical about the origin

**Figure 6.5**

Root positions to generate even polynomials: A , B, C, or any combination

|       |                             |                              |                             |
|-------|-----------------------------|------------------------------|-----------------------------|
| $s^5$ | 1                           | 6                            | 8                           |
| $s^4$ | <del>7</del> 1              | <del>42</del> 6              | <del>56</del> 8             |
| $s^3$ | <del>0</del> <del>4</del> 1 | <del>0</del> <del>12</del> 3 | <del>0</del> <del>0</del> 0 |
| $s^2$ | 3                           | 8                            | 0                           |
| $s^1$ | $\frac{1}{3}$               | 0                            | 0                           |
| $s^0$ | 8                           | 0                            | 0                           |

Summary of pole locations for Example 6.4

**Polynomial**

| Location         | Even<br>(fourth-order) | Other<br>(first-order) | Total<br>(fifth-order) |
|------------------|------------------------|------------------------|------------------------|
| Right half-plane | 0                      | 0                      | 0                      |
| Left half-plane  | 0                      | 1                      | 1                      |
| $j\omega$        | 4                      | 0                      | 4                      |

poles at :  $-7, \pm 2j, \pm 1.414j$

# Example

$$T(s) = \frac{20}{s^8 + s^7 + 12s^6 + 22s^5 + 39s^4 + 59s^3 + 48s^2 + 38s + 20}$$

$$s^8 \quad 1 \quad 12 \quad 39 \quad 48 \quad 20$$

$$s^7 \quad 1 \quad 22 \quad 59 \quad 38$$

$$s^6 \quad -\frac{(22-12)}{1} = -10 = -1 \quad -\frac{(59-39)}{1} = -20 \quad -\frac{(38-48)}{1} = 10 \quad -\frac{(0-20)}{1} = 20$$

$$s^5 \quad -\frac{(-2+22)}{-1} = 20 = 1 \quad -\frac{(1+59)}{-1} = 60 = 3 \quad -\frac{(2+38)}{-1} = 40 = 2$$

$$s^4 \quad -\frac{(-3+2)}{1} = 1 \quad -(-2-1)3 = 9 \quad -(0-2) = 2$$

$$s^3 \quad \emptyset \quad 2 \quad \emptyset \quad 3 \quad \emptyset \quad 0$$

$$s^2 \quad -\frac{(2-6)}{2} = 2 \quad -\frac{(0-4)}{2} = 2$$

$$s \quad -\frac{(4-\frac{9}{2})}{\frac{3}{2}} = -\frac{(\frac{1}{2})}{\frac{3}{2}} = \frac{1}{3}$$

$$s^0 \quad -\frac{(0-\frac{2}{3})}{\frac{1}{3}} = 2$$

→ 2 Roots in LHP

→ 2 Roots in RHP

→ 4 Roots on imaginary axis

$$P(s) = s^4 + 3s^2 + 2$$

$$\frac{dP(s)}{ds} = 4s^3 + 6s$$

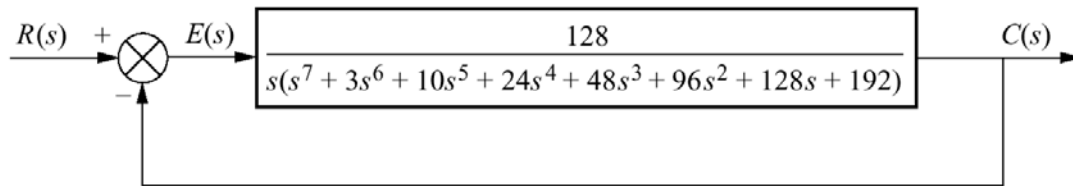


$$\begin{array}{ll}
 s^8 & 1 \\
 s^7 & 1 \\
 s^6 & -\frac{(22-12)}{1} = -10 = -1 \\
 s^5 & -\frac{(-2+22)}{-1} = 20 = 1 \\
 s^4 & -\frac{(-3+2)}{1} = 1 \\
 s^3 & 0 \quad 4 \quad 2 \\
 s^2 & -\frac{(3-6)}{2} = \frac{3}{2} \\
 s & -\frac{(4-\frac{9}{2})}{\frac{3}{2}} = -\left(\frac{-\frac{1}{2}}{\frac{3}{2}}\right) = \frac{1}{3} \\
 s^0 & -\frac{(0-\frac{2}{3})}{\frac{1}{3}} = 2
 \end{array}$$

**TABLE 6.9** Summary of pole locations for Example 6.5

| Location         | Polynomial             |                         |                         |
|------------------|------------------------|-------------------------|-------------------------|
|                  | Even<br>(fourth-order) | Other<br>(fourth-order) | Total<br>(eighth-order) |
| Right half-plane | 0                      | 2                       | 2                       |
| Left half-plane  | 0                      | 2                       | 2                       |
| $j\omega$        | 4                      | 0                       | 4                       |

**Example 6.8:** Find the number of poles in the left half-plane, the right half-plane, and on the  $j\omega$  axis for the system shown below. Draw conclusions about the stability of the closed-loop system.



$$T(s) = \frac{128}{s^8 + 3s^7 + 10s^6 + 24s^5 + 48s^4 + 96s^3 + 128s^2 + 192s + 128}$$

$$P(s) = s^6 + 8s^4 + 32s^2 + 64$$

$$\frac{dP(s)}{ds} = 6s^5 + 32s^3 + 64s + 0$$

|       |                             |                               |                               |                             |     |
|-------|-----------------------------|-------------------------------|-------------------------------|-----------------------------|-----|
| $s^8$ | 1                           | 10                            | 48                            | 128                         | 128 |
| $s^7$ | <del>3</del> 1              | <del>24</del> 8               | <del>96</del> 32              | <del>192</del> 64           |     |
| $s^6$ | <del>2</del> 1              | <del>16</del> 8               | <del>64</del> 32              | <del>128</del> 64           |     |
| $s^5$ | <del>0</del> <del>6</del> 3 | <del>0</del> <del>32</del> 16 | <del>0</del> <del>64</del> 32 | <del>0</del> <del>0</del> 0 |     |
| $s^4$ | <del>8</del> /<br>3 1       | <del>64</del> /<br>3 8        | <del>64</del> 24              |                             |     |
| $s^3$ | <del>-8</del> -1            | <del>-40</del> -5             |                               |                             |     |
| $s^2$ | <del>3</del> 1              | <del>24</del> 8               |                               |                             |     |
| $s^1$ | 3                           |                               |                               |                             |     |
| $s^0$ | 8                           |                               |                               |                             |     |

|       |                             |                               |                               |                             |     |
|-------|-----------------------------|-------------------------------|-------------------------------|-----------------------------|-----|
| $s^8$ | 1                           | 10                            | 48                            | 128                         | 128 |
| $s^7$ | <del>3</del> 1              | <del>24</del> 8               | <del>96</del> 32              | <del>192</del> 64           |     |
| $s^6$ | <del>2</del> 1              | <del>16</del> 8               | <del>64</del> 32              | <del>128</del> 64           |     |
| $s^5$ | <del>0</del> <del>6</del> 3 | <del>0</del> <del>32</del> 16 | <del>0</del> <del>64</del> 32 | <del>0</del> <del>0</del> 0 |     |
| $s^4$ | $\frac{8}{3}$ 1             | $\frac{64}{3}$ 8              | <del>64</del> 24              |                             |     |
| $s^3$ | <del>-8</del> -1            | <del>-40</del> -5             |                               |                             |     |
| $s^2$ | <del>3</del> 1              | <del>24</del> 8               |                               |                             |     |
| $s^1$ | 3                           |                               |                               |                             |     |
| $s^0$ | 8                           |                               |                               |                             |     |

**Table 6.14** Summary of pole locations for Example 6.8

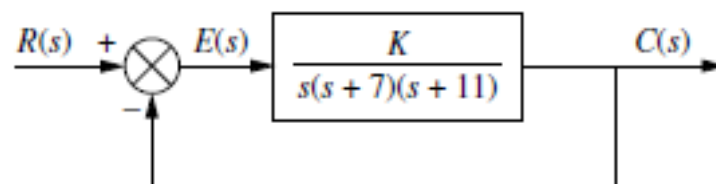
| Polynomial       |                       |                         |                         |
|------------------|-----------------------|-------------------------|-------------------------|
| Location         | Even<br>(sixth-order) | Other<br>(second-order) | Total<br>(eighth-order) |
| Right half-plane | 2                     | 0                       | 2                       |
| Left half-plane  | 2                     | 2                       | 4                       |
| $j\omega$        | 2                     | 0                       | 2                       |

Note: rhp = right half-plane, lhp = left half-plane.

## Stability Design via Routh-Hurwitz

**PROBLEM:** Find the range of gain,  $K$ , for the system of Figure 6.10 that will cause the system to be stable, unstable, and marginally stable. Assume  $K > 0$ .

**FIGURE 6.10** Feedback control system for Example 6.9



**SOLUTION:** First find the closed-loop transfer function as

$$T(s) = \frac{K}{s^3 + 18s^2 + 77s + K} \quad (6.20)$$

Next form the Routh table shown as Table 6.15.

**TABLE 6.15** Routh table for Example 6.9

|       |                       |     |
|-------|-----------------------|-----|
| $s^3$ | 1                     | 77  |
| $s^2$ | 18                    | $K$ |
| $s^1$ | $\frac{1386 - K}{18}$ |     |
| $s^0$ | $K$                   |     |

Since  $K$  is assumed positive, we see that all elements in the first column are always positive except the  $s^1$  row. This entry can be positive, zero, or negative, depending upon the value of  $K$ . If  $K < 1386$ , all terms in the first column will be positive, and since there are no sign changes, the system will have three poles in the left half-plane and be *stable*.

If  $K > 1386$ , the  $s^1$  term in the first column is negative. There are two sign changes, indicating that the system has two right-half-plane poles and one left-half-plane pole, which makes the system *unstable*.

If  $K = 1386$ , we have an entire row of zeros, which could signify  $j\omega$  poles. Returning to the  $s^2$  row and replacing  $K$  with 1386, we form the even polynomial

$$P(s) = 18s^2 + 1386 \quad (6.21)$$

Differentiating with respect to  $s$ , we have

$$\frac{dP(s)}{ds} = 36s + 0 \quad (6.22)$$

Replacing the row of zeros with the coefficients of Eq. (6.22), we obtain the Routh-Hurwitz table shown as Table 6.16 for the case of  $K = 1386$ .

**TABLE 6.16** Routh table for Example 6.9 with  $K = 1386$

|       |                  |      |
|-------|------------------|------|
| $s^3$ | 1                | 77   |
| $s^2$ | 18               | 1386 |
| $s^1$ | $\cancel{-0}$ 36 |      |
| $s^0$ | 1386             |      |

Since there are no sign changes from the even polynomial ( $s^2$  row) down to the bottom of the table, the even polynomial has its two roots on the  $j\omega$ -axis of unit multiplicity. Since there are no sign changes above the even polynomial, the remaining root is in the left half-plane. Therefore the system is *marginally stable*.

**PROBLEM:** For a unity feedback system with the forward transfer function

$$G(s) = \frac{K(s + 20)}{s(s + 2)(s + 3)}$$

find the range of  $K$  to make the system stable.

**PROBLEM:** For a unity feedback system with the forward transfer function

$$G(s) = \frac{K(s+20)}{s(s+2)(s+3)}$$

find the range of  $K$  to make the system stable.

$$T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + H(s)G(s)}$$

$$G(s) = \frac{K(s+20)}{s(s+2)(s+3)}$$

$$T(s) = \frac{K(s+20)}{s(s+2)(s+3) + K(s+20)} = \frac{K(s+20)}{s^3 + 5s^2 + (K+6)s + 20K}$$

$$s^3 \quad 1 \quad K+6$$

$$s^2 \quad 5 \quad 20K$$

$$s \quad -(3K-6)$$

$$s^0 \quad 20K$$

$$\begin{aligned} & \frac{20K - 5(K+6)}{5} \\ &= \frac{20K - 5K - 30}{5} \\ &= 3K - 6 \end{aligned}$$



For the system to be stable

$$6-3K > 0 \quad \& \quad 20K > 0$$

$$6 > 3K \quad K > 0$$

$$3K < 6$$

$$K < 2$$

range of  $K$  for stability  $0 < K < 2$

For the system to be marginally stable, there must be row of zeros.

i.e

$$6-3K = 0$$

$$6 = 3K$$

$$K = 2$$



$$s^3 \quad 1 \quad 8$$

$$s^2 \quad 5 \quad 40$$

$$s \quad \cancel{0} 10$$

$$s^0 \quad 40$$

$$P(s) = 5s^2 + 40$$

$$\frac{dP(s)}{ds} = 10s$$

Solve even polynomial

$$5s^2 + 40 = 0$$

$$s^2 = -8$$

$$\Rightarrow s = \pm \sqrt{8}j$$

Given the unity feedback system  $G(s) = \frac{Ks(s+2)}{(s^2 - 4s + 8)(s+3)}$

- a. Find the range of  $K$  for stability.
- b. Find the frequency of oscillation when the system is marginally stable.

Given the unity feedback system  $G(s) = \frac{Ks(s+2)}{(s^2 - 4s + 8)(s+3)}$

a. Find the range of  $K$  for stability.

b. Find the frequency of oscillation when the system is marginally stable.

$$G(s) = \frac{Ks(s+2)}{(s^2 - 4s + 8)(s+3)}$$

$$\begin{aligned} T(s) &= \frac{Ks(s+2)}{(s^2 - 4s + 8)(s+3) + Ks(s+2)} \\ &= \frac{Ks^2 + 2Ks}{s^3 + (K-1)s^2 + (2K-4)s + 24} \end{aligned}$$

Routh Table

|       |                              |        |
|-------|------------------------------|--------|
| $s^3$ | 1                            | $2K-4$ |
| $s^2$ | $K-1$                        | 24     |
| $s$   | $\frac{2K^2 - 6K - 20}{K-1}$ |        |
| $s^0$ | 24                           |        |

$$\begin{aligned} & - \left[ \frac{24(1) - (K-1)(2K-4)}{K-1} \right] \\ &= - \left[ \frac{24 - (2K^2 - 4K - 2K + 4)}{K-1} \right] \\ &= - \left[ \frac{24 - 2K^2 + 6K - 4}{K-1} \right] \\ &= \frac{2K^2 - 6K - 20}{K-1} \end{aligned}$$

For the system to be stable

$$k-1 > 0 \quad \& \quad \frac{2k^2-6k-20}{k-1} > 0$$

$$k > 1 \quad \& \quad 2k^2-6k-20 > 0$$

$$k^2-3k-10 > 0$$

$$k^2-5k+2k-10 > 0$$

$$k(k-5)+2(k-5) > 0$$

$$(k+2)(k-5) > 0$$

$$k > 1$$

$$k > -2$$

$$k > 5$$

IF  $\boxed{k > 5}$  then system is stable.

For the system to be marginally stable

$$\frac{2k^2 - 6k - 20}{k-1} = 0$$

$$2k^2 - 6k - 20 = 0$$

$$(k+2)(k-5) = 0$$

$$k = -2 \quad k = 5$$

$k$  can't be negative, so take  $k = 5$ .

|       |                |    |
|-------|----------------|----|
| $s^3$ | 1              | 6  |
| $s^2$ | 4              | 24 |
| $s$   | <del>0</del> 8 |    |
| $s^0$ | 24             |    |

$$P(s) = 4s^2 + 24$$

$$\frac{dP(s)}{ds} = 8s$$

no. sign change  $\Rightarrow$  no roots in right half  $s$ -plane.  
even polynomial has imaginary roots.

$$4s^2 + 24 = 0$$

$$s^2 + 6 = 0$$

$$s = \pm \sqrt{6}j$$

# Stability in state space

Closed loop transfer function is

$$T(s) = \frac{Y(s)}{U(s)} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} + \mathbf{D}$$

where  $(s\mathbf{I} - \mathbf{A})^{-1} = \frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})}$

$\det(s\mathbf{I} - \mathbf{A})$  Is the denominator of transfer function.

$\det(s\mathbf{I} - \mathbf{A}) = 0$  is characteristic equation and gives location of poles.

Values of the system's poles are equal to the eigenvalues of the system matrix  $\mathbf{A}$

# Stability in state space

Stability of a system in state space can be found by calculating the Eigen values of matrix  $A$ .

All the Eigen values must be in left half s-plane.

Procedure:

- Using state space matrices first form  $(sI - A)$
- Find  $\det(sI - A)$
- Using this polynomial(characteristic equation), Generate and interpret Routh table

**PROBLEM:** Given the system

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 3 & 1 \\ 2 & 8 & 1 \\ -10 & -5 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u \quad (6.33a)$$

$$y = [1 \ 0 \ 0] \mathbf{x} \quad (6.33b)$$

find out how many poles are in the left half-plane, in the right half-plane, and on the  $j\omega$ -axis.

**SOLUTION:** First form  $(s\mathbf{I} - \mathbf{A})$ :

$$(s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 3 & 1 \\ 2 & 8 & 1 \\ -10 & -5 & -2 \end{bmatrix} = \begin{bmatrix} s & -3 & -1 \\ -2 & s-8 & -1 \\ 10 & 5 & s+2 \end{bmatrix} \quad (6.34)$$

Now find the  $\det(s\mathbf{I} - \mathbf{A})$ :

$$\det(s\mathbf{I} - \mathbf{A}) = s^3 - 6s^2 - 7s - 52 \quad (6.35)$$

Using this polynomial, form the Routh table of Table 6.18.

**TABLE 6.18** Routh table for Example 6.11

|       |                                      |     |                |     |
|-------|--------------------------------------|-----|----------------|-----|
| $s^3$ |                                      | 1   |                | -7  |
| $s^2$ | <del>-6</del>                        | -3  | <del>-52</del> | -26 |
| $s^1$ | <del><math>\frac{47}{3}</math></del> | -1  | <del>0</del>   | 0   |
| $s^0$ |                                      | -26 |                |     |